



The necessity of the €STR rate

Benchmark Rates Reform - Marcus Evans conferences
London, 15th March 2019

Agenda

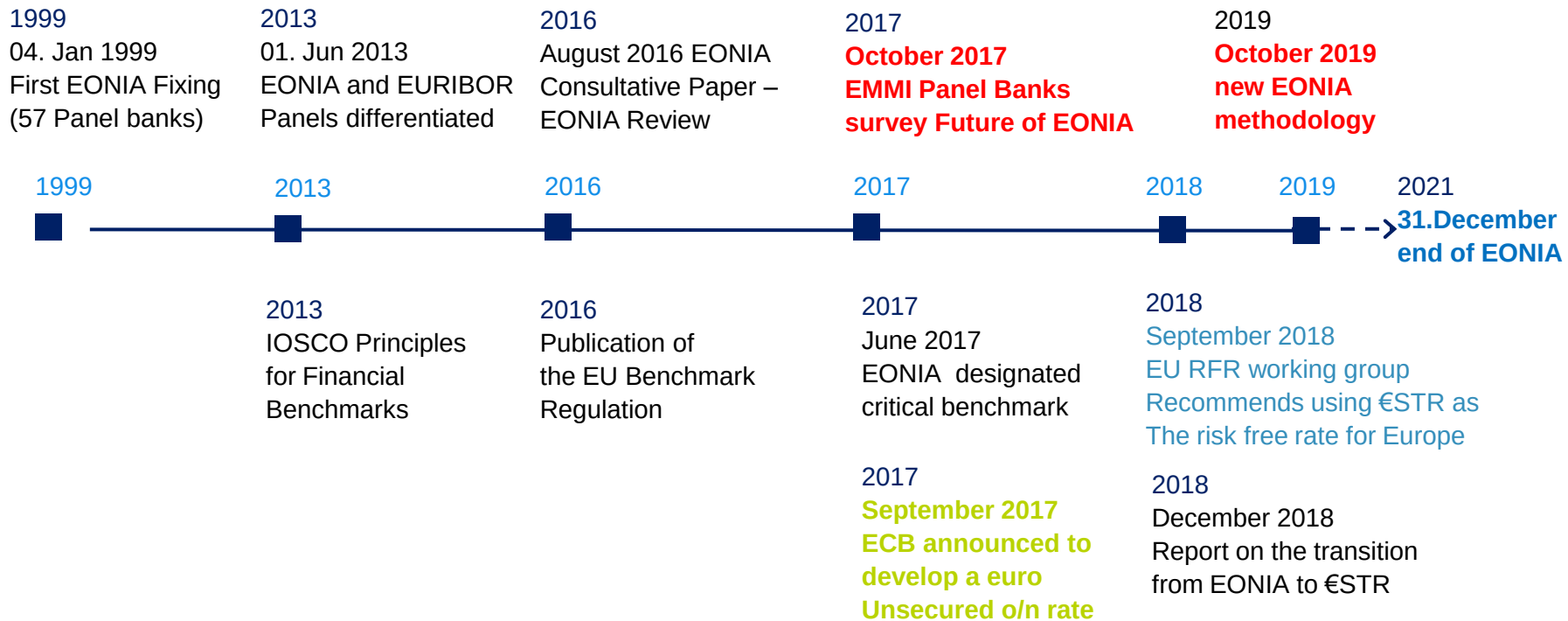
- › Why is the EONIA rate no longer robust?
- › The consequential differences between Eonia and €STR
- › Historical data and the 50 MMSR sample
- › How will the €STR and Eonia rates change over time, how will this impact the market?

Why is the EONIA no longer robust?



EONIA History

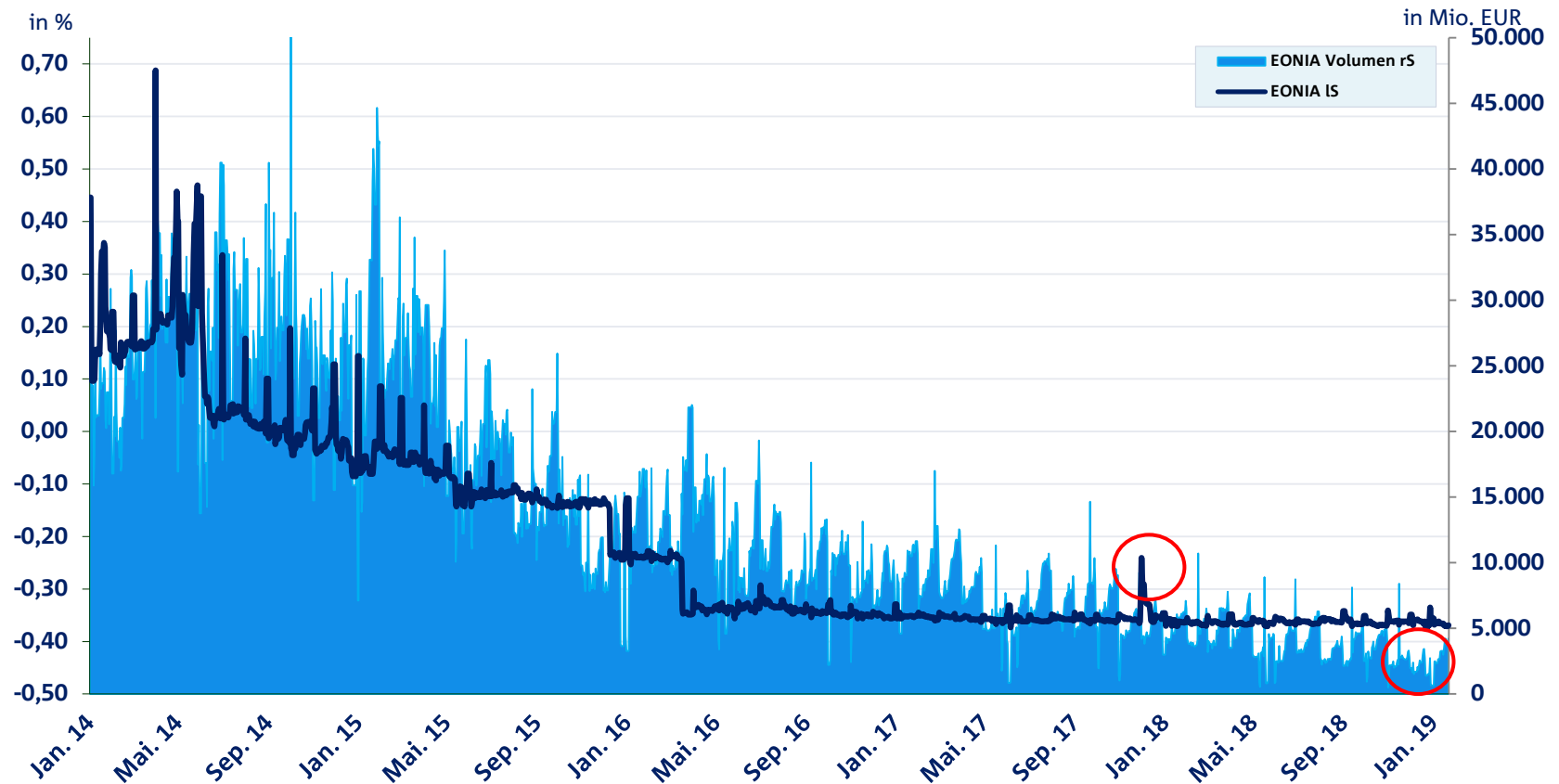
EMMI



Euribor / Eonia panel banks

	Euribor Panel (19)	Eonia Panel (28)	Panel 2011 (44)
Austria		Erste Group Bank AG	Raiffeisen Zentralbank (2013)
Belgium	Belfius		KBC (2014)
Finland		Nordea - Pohjola	
France	BNP-Paribas - HSBC France - Natixis/BPCE - Crédit Agricole s.a - Société Générale	Crédit Industriel et Commercial CIC	Banque Postale (2014)
Germany	Deutsche Bank - DZ Bank	BayernLB - Nord LB - LBBW - HeLaBa	Commerzbank (2014) - Landesbank Berlin (2013) - WestLB
Greece	National Bank of Greece		
Ireland		Bank of Ireland	Allied Irish (2013)
Italy	Intesa Sanpaolo - UniCredit	- Monte dei Paschi di Siena (2019)	UBI Banca (2014)
Luxembourg	Banque et Caisse d'Épargne de l'État		
Netherlands	ING Bank		Rabo Bank (2013)
Portugal	Caixa Geral De Depósitos (CGD)		
Spain	Banco Bilbao Vizcaya Argentaria - Banco Santander - CECABANK - CaixaBank S.A.		
Other EU Banks	Barclays		Danske Bank (2015) Royal Bank of Scotland (2013) Svenska (2013)
International Banks			London Branch of JP Morgan Chase Bank N.A (2016) - Bank of Tokyo Mitsubishi (2016) – Citibank (2012) – UBS (2013)
Source:EMMI	Red = MMSR bank		

EONIA rate & volume



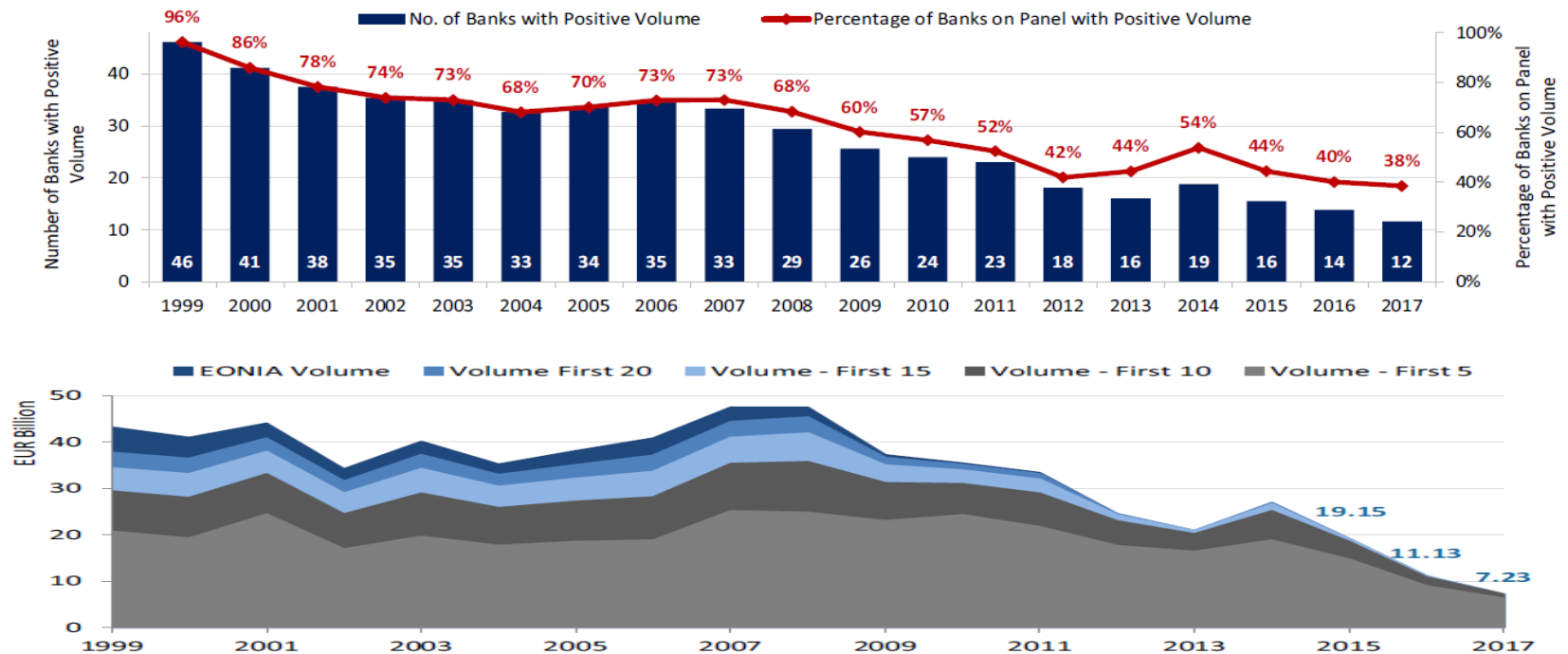
Source: ThomsonReuters/ECB



Outlier transactions and regional public holidays have a big impact on EONIA volume and interest rate = not compliant with the Benchmark Regulation

EONIA trend

➔ Panel Banks with non-zero submissions

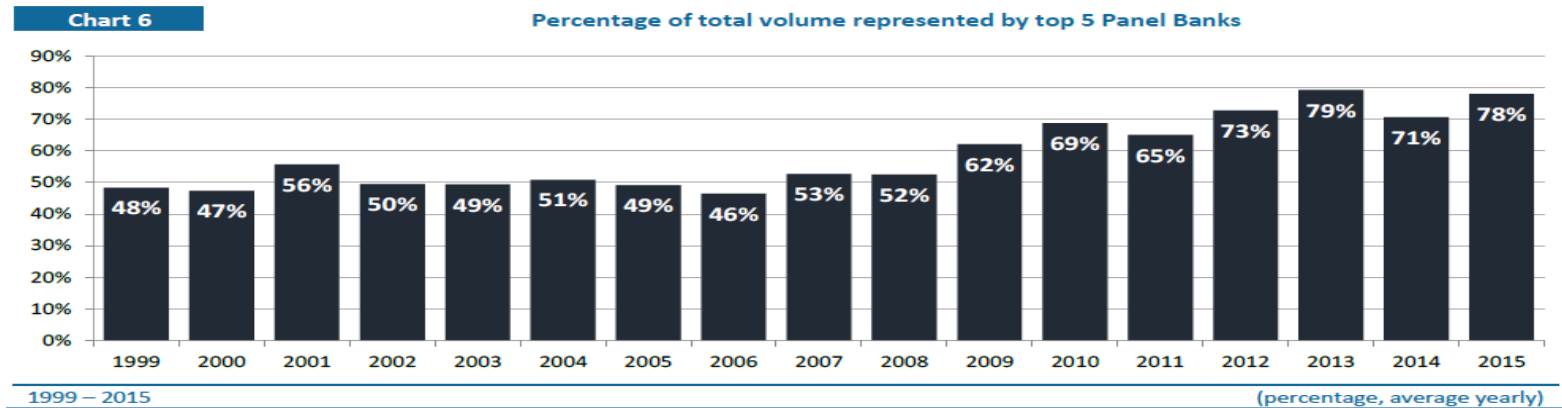


Source: EMMI

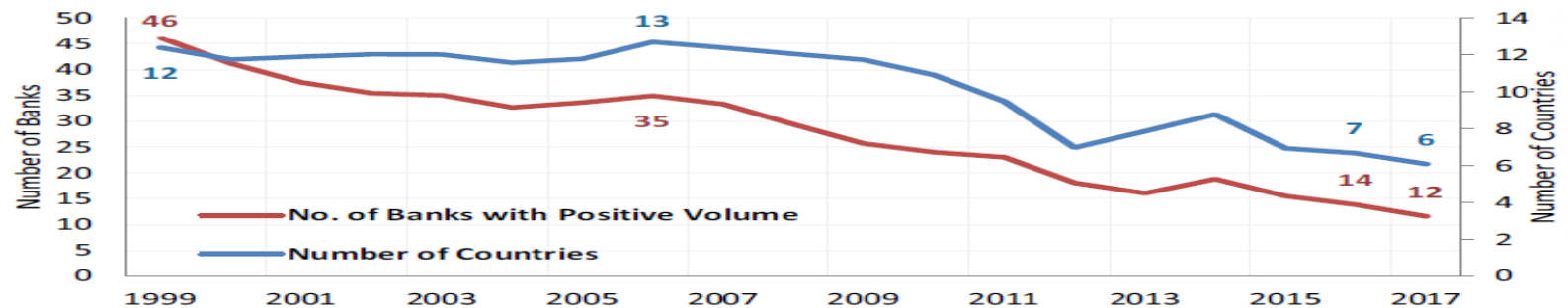


Monetary policy and regulation encouraged the concentration in EONIA

EONIA trend



➔ Geographic concentration indicators



Source: EMMI



Too concentrated: transactions of only a few banks from a very small number of countries

The consequential differences between EONIA and €STR



€STR – Euro short term rate

1.

Announced 09/2017
Final methodology 06/2018
Daily provision of €STR
from **10/2019**

2.

The ECB uses data
from the money market
statistical report (MMSR)
50 banks

3.

Deposits with financial
counterparties (volume
average EUR 30 bn.)
Historical data available
since **06/2016**

4.

According to ECB calculations
€STR will currently be around
4 bp below the deposit facility
(= -0.44%)

5.

09/2018 The EU working group
recommends using €STR as
the risk free rate for Europe



Conclusion: as is so often the case, the ECB is the lender of last resort

Comparison of EONIA/€STR

Name	EONIA (European overnight index average)	€STR (European short term rate)
Administrator	EMMI (European Money Market Institute)	ECB
Definition	Euro o/n unsecured lending interbank transaction	Euro o/n unsecured borrowing wholesale transactions
Data basis	28 panel banks	50-bank MMSR (Money Market Statistical Report)
Publication	Same day around 7:00 p.m. (from Q4 2019 next day by 9:00 a.m)	Next day by 9:00 a.m. (from Q4 2019)
Published data	› Interest rate (3 decimal places) › Volume EUR m	› Interest rate (3 decimal places) › Volume EUR m › Number of reporting banks (before trimming) › Number of transactions (before trimming) › % share of the five largest supplier banks › Calculation methodology (normal/emergency)
Methodology	No limit (rounded to EUR m)	From EUR 1 m mean trimming 25%
Volume	EUR 1 bn to 5 bn	EUR 30 bn
Benchmark regulation	Non-compliant (no order received from panel banks). Theoretical grandfathering is possible even after 12/19	Compliant
Business continuity plan	If 4 or fewer banks report transactions	If less than 20 banks conduct transactions or 5 banks combine to provide more than 75% of the volume
Fixing level determined:	Approx. -0.36% to -0,37%	Approx. -0.44% to -0,46%
Historical data	Since 01/99	Since 06/16



Historical data and the 50 bank MMSR sample

50 MMSR Banks

ABN AMRO Bank N.V.	Caisse Fédérale de Crédit Mutuel	ING-DiBa AG
Allied Irish Banks plc	CaixaBank, S.A. EONIA	Intesa Sanpaolo S.p.A. EONIA
Banca IMI S.p.A.	Cassa Depositi e Prestiti Societa' per Azioni	KBC Bank NV
Banca Monte dei Paschi di Siena S.p.A. EONIA	Commerzbank Aktiengesellschaft	Kreditanstalt für Wiederaufbau
Banco Bilbao Vizcaya Argentaria, S.A. EONIA	Coöperatieve Rabobank U.A.	La Banque Postale
Banco de Sabadell, S.A.	Crédit Agricole Corporate and Investment Bank	Landesbank Baden-Württemberg EONIA
Banco BPM Societa' per Azioni	Crédit Agricole S.A.	Landesbank Hessen-Thüringen Girozentrale EONIA
Banco Santander, S.A. EONIA	Crédit Lyonnais	Natixis EONIA
Bankia, S.A.	DekaBank Deutsche Girozentrale	Norddeutsche Landesbank -Girozentrale- EONIA
Banque fédérative du crédit mutuel	Deutsche Bank Aktiengesellschaft EONIA	Nordea Bank Abp EONIA
Bayerische Landesbank EONIA	DB Privat- und Firmenkundenbank AG	NRW.BANK
Belfius Banque SA EONIA	Dexia crédit local	Piraeus Bank, S.A.
BNG Bank N.V.	DZ Bank AG Deutsche Zentral-Genossenschaftsbank EONIA	Société Générale EONIA
BNP Paribas EONIA	Hamburg Commercial Bank AG	UniCredit Bank AG
BNP Paribas Fortis SA	HSBC France EONIA	UniCredit Bank Austria AG
BPCE	ING Bank N.V. EONIA	UniCredit, Societa' per Azioni EONIA
Caisse des dépôts et consignations - section générale	ING Belgique SA	

Source: ECB/EMMI



50 MMSR includes 19 EONIA Panelbanks, but only 10 out of 19 EURO countries (EONIA 13 different countries)

Pre-€STR vs. €STR

What is the difference between pre-ESTER and ESTER?

Pre-ESTER is calculated using the same methods as defined for ESTER. Pre-ESTER differs in that it is based on final data and includes all revisions in terms of cancellations, corrections and amendments submitted by reporting agents at the time of calculating the rate, while ESTER will be published every morning and take into account only the data received by the submission deadline of 07:00 CET that morning.

Disclaimer

Pre-ESTER is only a set of indicators for ESTER. Its publication is for information purposes only and the data are not intended for use as a benchmark in any market transaction, whether directly or indirectly.



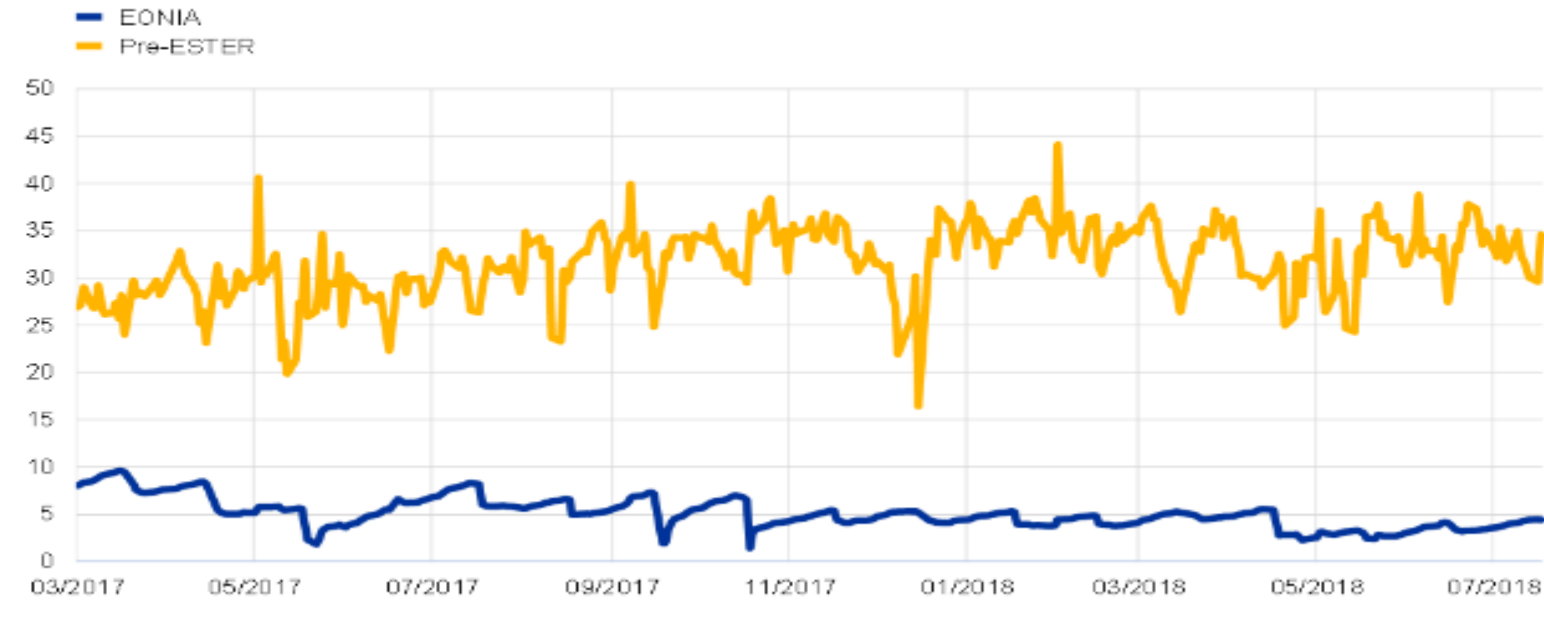
Starting as of January 2019 there is a shadow production of €STR

A period of live testing of all procedures and systems without publication

Pre-€STR very similar to €STR

Volume of €STR vs. EONIA

(EUR billion)

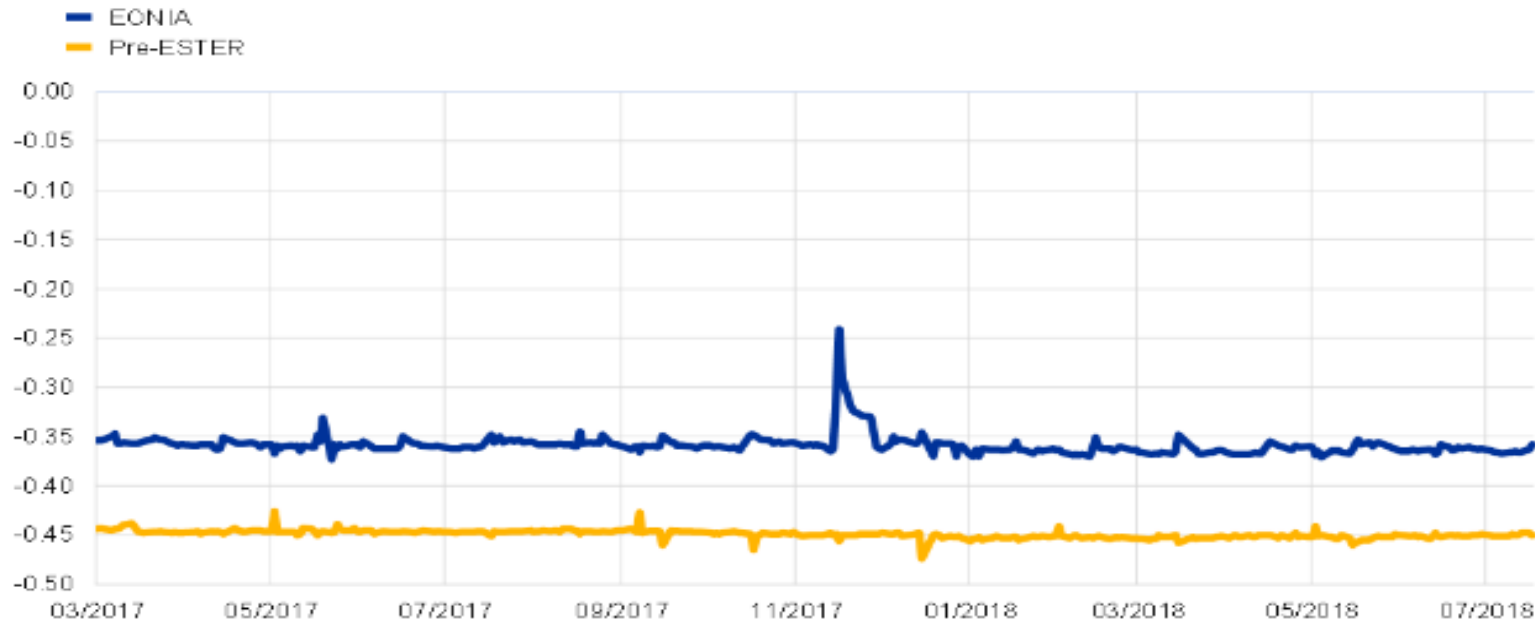


Source: ECB/EMMI



€STR provides much higher volume while also exhibiting lower concentration than with EONIA

Volatility of €STR vs. EONIA

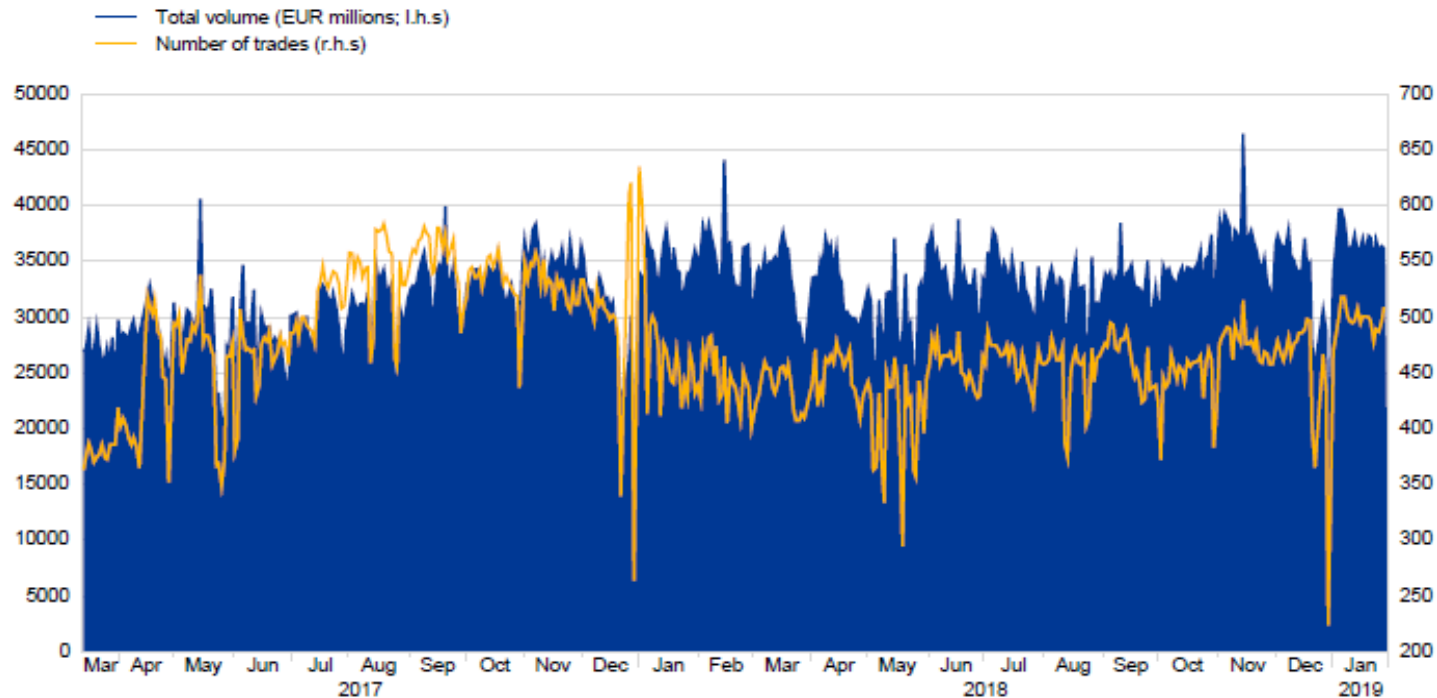


Source: ECB/EMMI



The trimming method (25% top/bottom) with €STR reduces considerably volatility
€STR is quite stable

€STR - Daily total volume and number of trades



Source: ECB



No contingency triggered by lack of transactions

€STR - Data sufficiency

Table 1

Data sufficiency parameters for three groups of counterparties

Measure	(1) Deposit-taking financial corporations	(2) Financial corporations	(3) Financial and non-financial corporations
Average number of banks	23	31	33
Lowest number of banks	15	24	26
Largest number of banks	29	36	38
Average number of countries	9	10	10
Lowest number of countries	6	7	7
Largest number of countries	10	10	10
Average number of transactions	266	592	688
Lowest number of transactions	70	202	277
Largest number of transactions	770	1114	1198
Average daily volume (EUR billions)	8.3	29.8	39.0
Lowest daily volume (EUR billions)	1.9	6.8	15.1
Largest daily volume (EUR billions)	19.0	41.5	53.6

Note: This table is based on daily MMSR data and covers the period from 1 August 2016 to 15 January 2018, capturing unsecured overnight transactions in the form of fixed-rate deposits.

Source: ECB/RFR working group

Measure	Average	Highest	Lowest
Five largest banks' share of total daily volumes	59%	74%	50%
Six largest banks' share of total daily volumes	64%	78%	56%
Belgium's share of total daily volumes	15%	22%	5%
Germany's share of total daily volumes	22%	33%	14%
Spain's share of total daily volumes	5%	9%	1%
France's share of total daily volumes	36%	55%	25%
Italy's share of total daily volumes	1%	5%	0%
Netherlands' share of total daily volumes	17%	32%	8%
Others' share of total daily volumes	4%	8%	1%

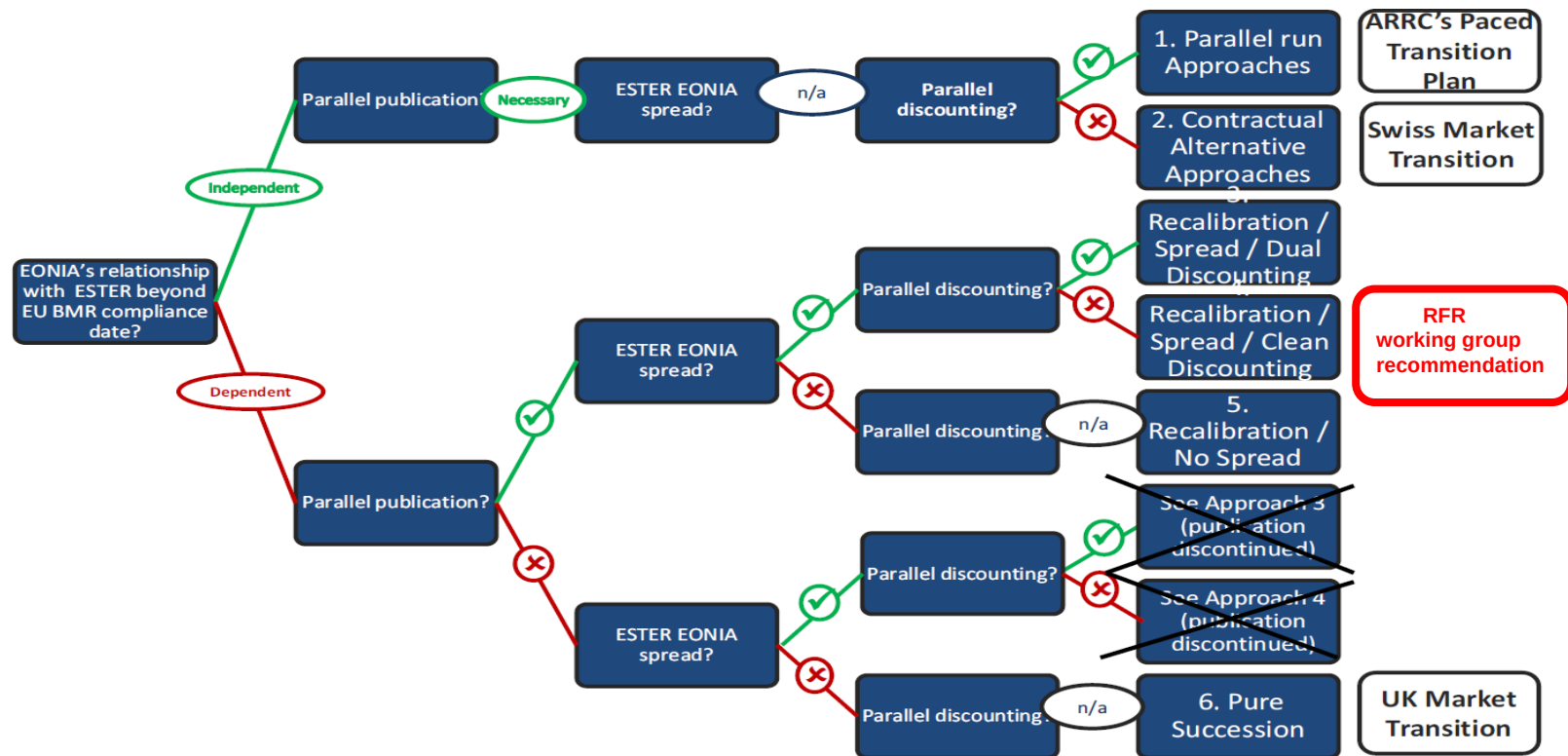


ECB puts a lot of effort in transparency of €STR

How will the €STR
and Eonia rates
change over time, how
will this impact the
market?



Many paths lead from EONIA to €STR



Source: ECB/RFR working group



There is no model solution for seamless transition

Feedback on transition EONIA to €STR

Chart 3

Extent of agreement with the working group's recommended transition path from EONIA to ESTER

February 2019

(number of responses)

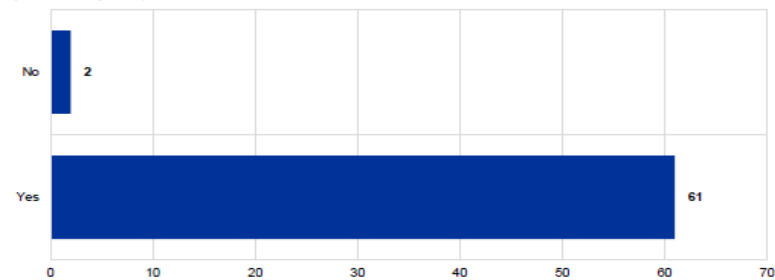


Chart 7

Extent of support for the authorisation and supervision of the recalibrated EONIA

February 2019

(number of responses)

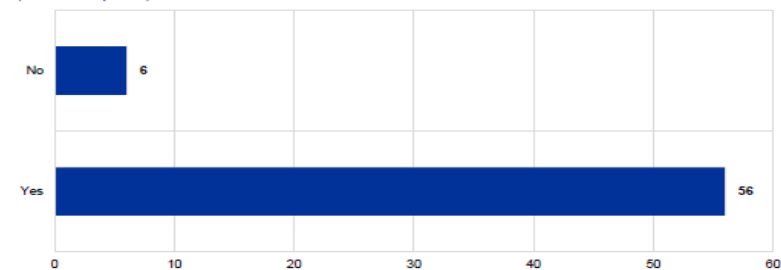


Chart 5

Extent of agreement with the recalibrated EONIA deadline of end-2021

February 2019

(number of responses)

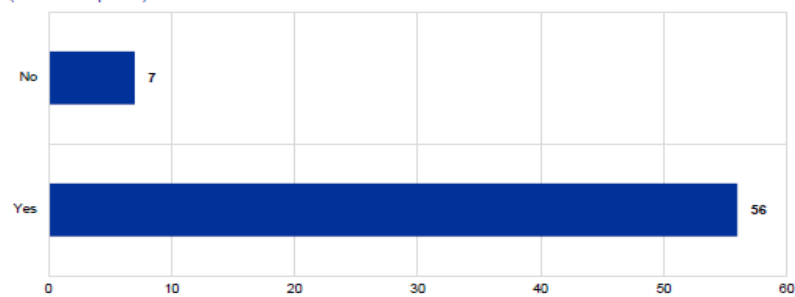
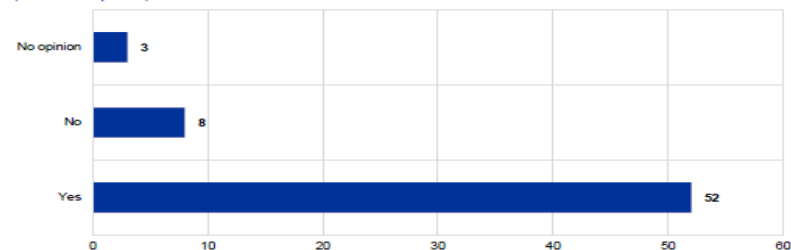


Chart 8

Extent of agreement with the working group's proposal of a spread methodology based on a trimmed mean (15% 1Y)

February 2019

(number of responses)



Source: ECB/RFR working group



Respondents (97%) broadly agreed with the working group's recommendation

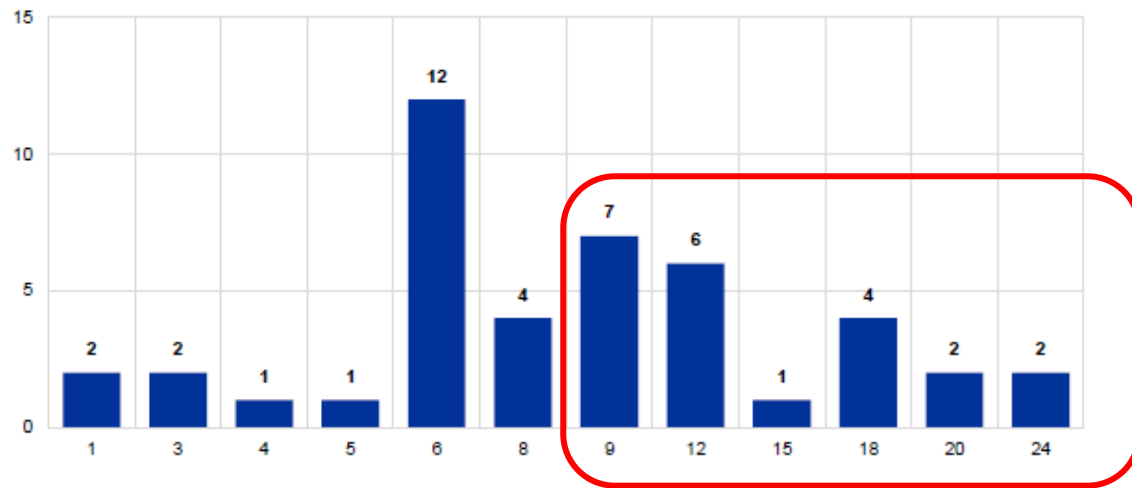
€STR T+1 publication

Chart 10

Number of months required to make systems ready for ESTER T+1 publication

February 2019

(number of responses)



Source: ECB Secretariat to the working group on euro risk-free rates.

Notes: The sum of the number of responses does not equal the total number of responses received as some respondents did not reply to this question.

Source: ECB/RFR working group

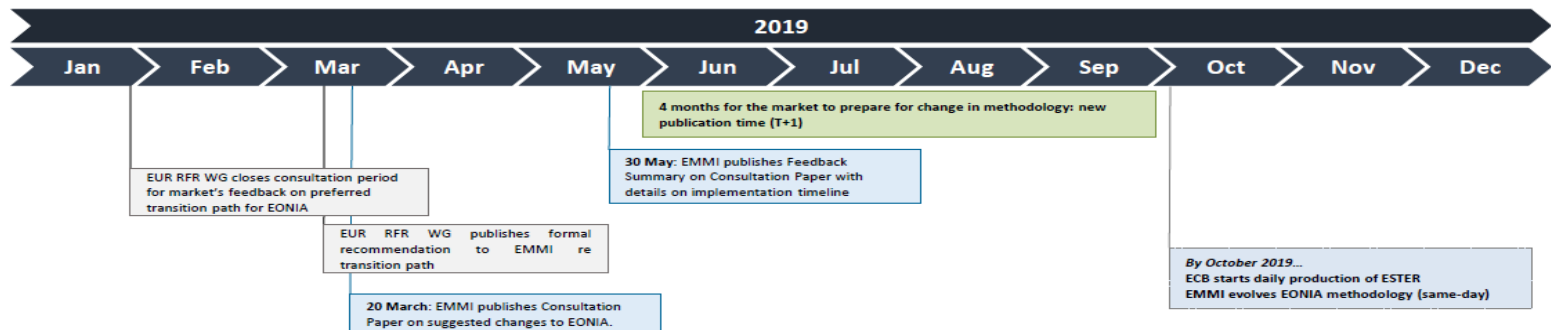


€STR is ready but not the banks.....looks like a slow €STR start

EMMIs EONIA Recalibration/Transition schedule



EONIA Recalibration/Transition



- ➡ Suggested transition path implies the redefinition of EONIA as ESTER plus a *spread*.
- ➡ Publication time will need to change, so that it follows ESTER's publication in the morning of T+1 (for trades executed throughout day T).
- ➡ It remains to be agreed which institution should be responsible for the calculation of the *spread* to be added to ESTER.

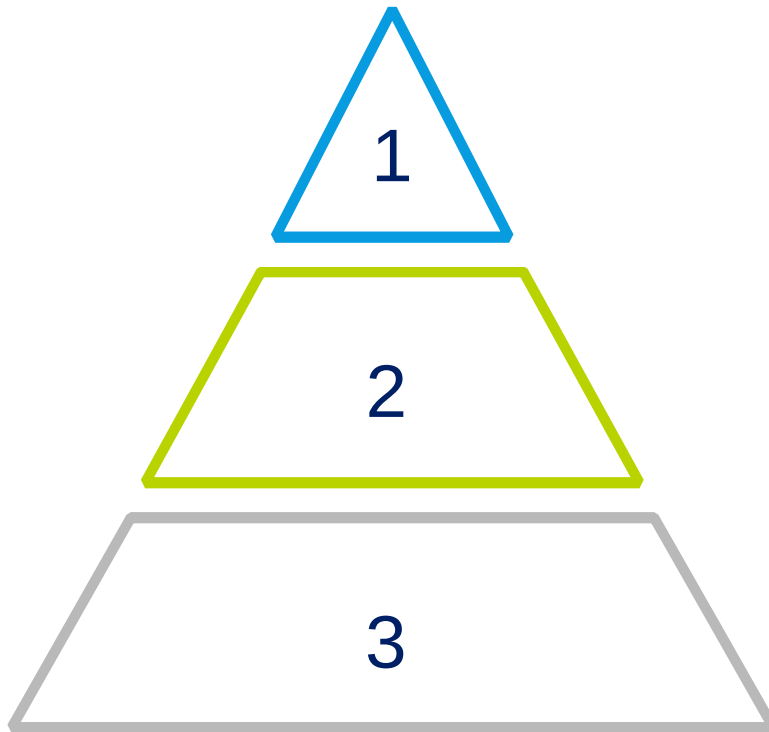


Working Group on euro risk-free rates | 27 February 2019

European Money Markets Institut



Conclusion



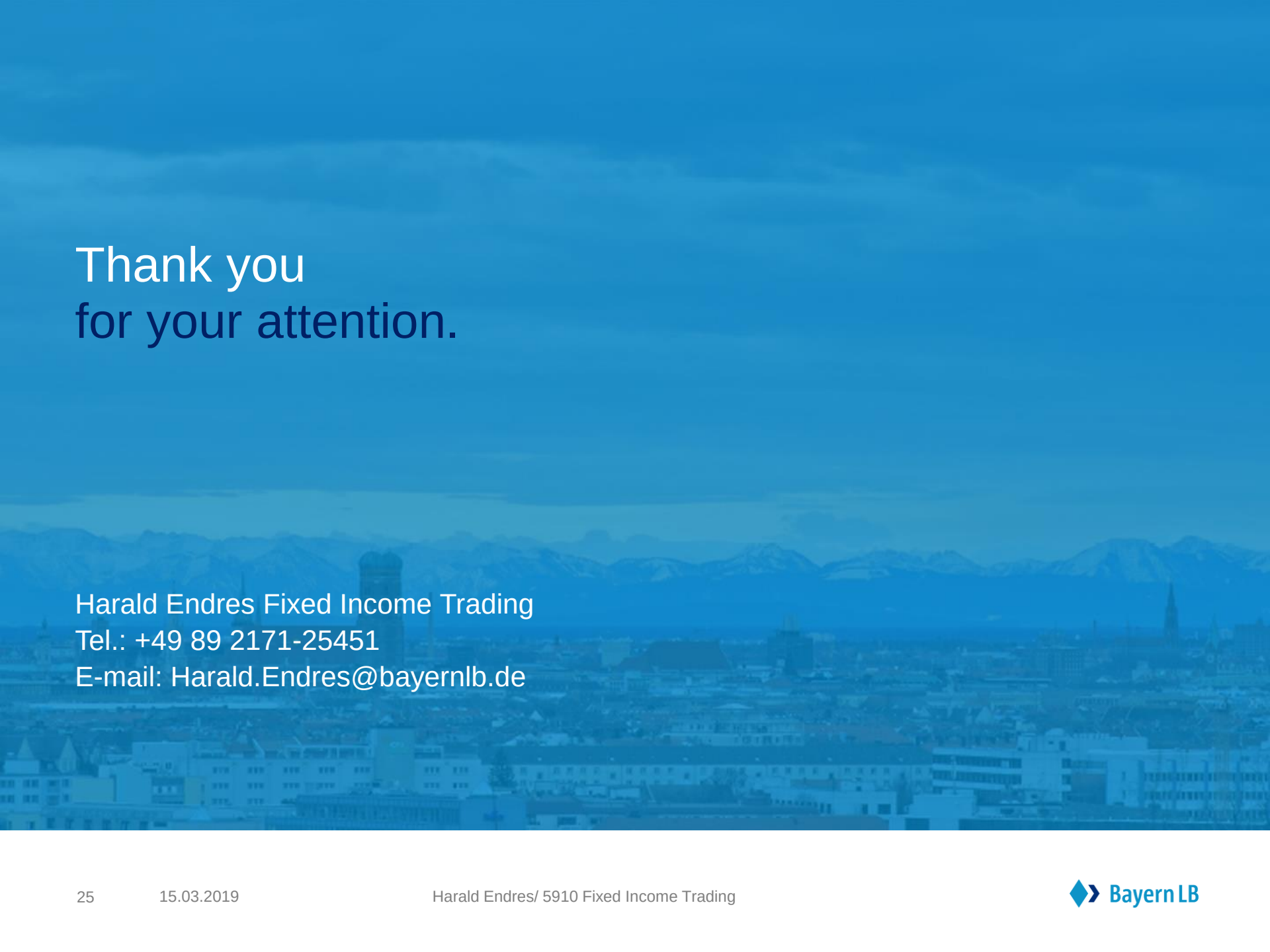
✓ **€STR**
Europe finally has an official risk free rate

⚠ **EONIA**
The end is near. EONIA will disappear completely by the end of 2021 at the latest

✓ **EURIBOR**
There is a good chance that EURIBOR will continue to be used for many years



Europe was late in tackling the benchmark issue but it is now close to finding solid solutions



Thank you
for your attention.

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We prepared these proposals for action with the utmost care. As agreed, you have only provided us with partial details of your financial situation and you have restricted the investment objectives accordingly. The proposals for action are therefore based only on the analysis of the portfolio data you have provided. As a result, it cannot be excluded that an alternative strategy may have been more suitable for you, had the full details of your financial situation been available.



The Bavarian bank for
the German economy.